

JOHN CARROLL

Forward Deployed Engineering | Enterprise Platforms | Agentic Systems

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AI architecture and engineering leader in The Cigna Group / Evernorth's AI Enablement Office, leading enterprise initiatives across agentic systems, retrieval, platform delivery, and business-aligned AI solutions.

PROFESSIONAL EXPERIENCE

Cigna | Evernorth (via Solomon Page) - Senior Principal Architect

Nov 2025 - Present | Atlanta, GA

- Lead applied AI architecture and forward deployed engineering for the AI Enablement Office, embedding with business units to identify operational problems, build AI prototypes, and convert high-value POCs into scalable enterprise platforms.
- Help establish the internal Forward Deployed Engineering model, including the FDE playbook, hiring guidance, delivery structure, and repeatable operating model for business-aligned AI solution delivery.
- Architect enterprise AI systems across agentic workflows, RAG, search/retrieval, analytics integrations, evaluation, and governance, translating business needs into secure, reliable, and production-ready technical designs.
- Guide engineering execution for an FDE pod by coordinating developers, setting technical direction, defining phases, managing roadmap tradeoffs, and aligning implementation commitments with stakeholder expectations.
- Design and implement retrieval pipelines using ingestion, chunking, indexing, grounding, semantic/hybrid search, evaluation, and quality monitoring to improve answer reliability, traceability, and enterprise reuse.
- Build production services and full-stack prototypes integrating LLM tool use, safety guardrails, measurable evaluation, ETL pipelines, knowledge graph-assisted retrieval, multimodal workflows, and translation capabilities.
- Support internal AI platform strategy and build-vs-buy decisions by evaluating enterprise AI patterns, reusable architecture, data governance needs, and scalable internal delivery models.
- Enable executive-facing AI adoption through senior leadership demos, stakeholder education, and delivery leadership for production AI systems.

Microsoft - Senior Solution Engineer | AI Apps & Agents Global Black Belt

Sep 2023 - Nov 2025 | Atlanta, GA

- Trusted advisor and architect for Fortune 500 clients, driving \$200M+ in AI-related revenue within my first year by shaping AI adoption strategies, leading technical deep dives, and unblocking enterprise-scale deployments for companies including Nvidia, Walmart, Coke, and Cisco.
- Engineer and deploy production-grade agentic AI systems that plan, reason, and take safe, reliable action across mission-critical workflows—balancing business requirements with technical execution.
- Designed and shipped retrieval-augmented generation (RAG) pipelines, including indexing and chunking strategies, semantic search, and grounding, which improved precision and recall for enterprise search and automation.

EDUCATION

Georgia Institute of Technology

M.S. Computer Science

AI Specialization

Expected Start: August 2026

Quantic School of Business and Technology

Executive MBA

March 2025 - Expected July 2026

Auburn University

B.S. Software Engineering

August 2011 - May 2016

CERTIFICATIONS

Neo4j Graph Data Science (Nov 2024)

Azure AI Engineer Associate (Dec 2025)

RECOGNITION

Lead Engineer - Cigna

Fastest Production Release

January 2026

Judge - Microsoft

Global Hackathon, AI Agents Category

August 2025

Lead Developer - Microsoft

AI Global Black Belt Hackathon

1st Place, April 2024

Team Lead - Boeing

Alabama 20th Annual STARS Award

"Predictability and Stability"

February 2023

- Built AI agents using LangChain, LangGraph, Semantic Kernel, AutoGen, and PromptFlow, implementing tool-use orchestration, memory, safety guardrails, and cost/latency tradeoff strategies to ensure enterprise-grade reliability.
- Leveraged evaluation frameworks and AI orchestration at the network level (Azure API Management / API Center AI Gateway) to optimize performance, safety, and efficient model utilization across high-scale deployments.
- Delivered end-to-end AI solutions (computer vision, OCR, NLP-to-SQL, video indexing) that automated decision-making, reduced operational costs, and accelerated digital transformation initiatives.
- Created reusable accelerators, POCs, and solution IP adopted by clients and Microsoft field teams, multiplying impact and standardizing patterns for scalable customer-facing AI deployments.

Boeing - Senior AI/ML Software Engineer

Jun 2019 - Sep 2023 | Huntsville, AL

- Spearheaded value stream data visualization and analytics solution focused on design and procurement processes, enhancing business intelligence for high-profile programs (EUS, SLS, ISS, PAC3, GMD). Delivered \$25M in annual efficiency savings, earning Alabama's 20th STARS Award.
- Headed the redesign of a 3D factory layout Unity app, optimizing UX and boosting operator efficiency, resulting in a 25% increase in conversion rates and a 15% reduction in user churn.
- Implemented a root cause analysis solution for part changes in engineering processes, driving key insights through metrics, substantially reducing subsystem errors for 1,000+ daily users.
- Engineered a Natural Language Processing (NLP)-based authoring tool for aircraft requirements, enhancing accuracy, efficiency, and reducing ambiguity, as part of an initiative aimed at achieving \$10B in annual cost savings across the enterprise by minimizing revision cycles.

Hexagon Safety & Infrastructure - Full-Stack Software Engineer

Jun 2016 - Jul 2019 | Madison, AL

- Engineered Smart Cities public safety SaaS on Azure, integrating IoT, robot/drone command-and-control, and computer vision-powered video analytics across real-time, scalable, multi-tenant architecture.
- Led DevSecOps execution for securing cloud resources and delivery pipelines; built and maintained CI/CD automation and infrastructure-as-code, improving reliability and deployment consistency.
- Supported compliance/quality standards (e.g., Microsoft Gold Partner expectations, ISO 9001 practices) through robust engineering processes and operational rigor.

ADDITIONAL EXPERIENCE

Auburn University - Computer Science Teaching Assistant *Aug 2015 - May 2016 | Auburn, AL*

Neptune Technology Group - Software Engineer (Co-op Program) *Jan 2013 - May 2015 | Tallahassee, AL*

(Completed three software/electrical engineering co-op rotations.)

Independent Consultant & Full-Stack Software Engineer *Jan 2011 - Present*

SKILLS

Core: Applied AI Engineering; Full-Stack Development; DevSecOps; AI Product Prototyping → Production; Data Analytics & Visualization; Agile Delivery (Scrum/Kanban)

Languages: Python; TypeScript/ JavaScript; C#; SQL; Node.js; Bash; Java

Applied AI Systems: LLM applications; RAG; agentic orchestration; tool/ function calling; prompt design; safety guardrails; evaluation/ benchmarks; quality monitoring; human-in-the-loop workflows

Retrieval & Search: chunking/indexing strategies; hybrid search; reranking; grounding/citations; query rewriting; caching; latency/cost tradeoffs

Data & Knowledge: knowledge graphs; entity linking; schema/ontology modeling; embeddings; vector databases; metadata pipelines; ETL/ELT; data quality checks

Backend & APIs: REST/JSON APIs; authn/authz patterns; service integration; background jobs/queues, testing strategies; performance tuning

Frontend: React; Next.js; Angular; Vue; component-driven UI; accessibility basics; user-centered design

Cloud/DevOps: Azure; AWS; GCP; Docker; Kubernetes; CI/CD; infrastructure-as-code; secrets management; observability

Frameworks/Tools: LangChain; LangGraph; LangSmith; Grafana; AutoGen; GitHub; Linear; Cursor, Claude Code; Codex